

EDA Notes Summary: Bike Retail Project 2

The first EDA analysis 5/5/2026

Dimensions Exploration

- There are 13 states and 20 cities from which our customers originate from (customer dim)
- There are 5 brands, 8 categories (5 accessories and 3 bikes), 60 models
- There are 3 sales channels: online, dealer, store
- We have 6 stores

From this analysis, I would like to further dig into the following:

1. See what states we have the most sales in
 2. See what stores are owned by us and which are distributors
 3. What bike categories sell the most
 4. What states have the most sales
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Date Exploration

First Order: Jan/1/2023

Last Order: March/6/2026

Month Range: 38

New Customers:

- 2022 - 38 new customers
- 2023- 42 new customers
- 2024- 35 new customers
- 2025- 5 new customers

Observations:

1. Why did we have 38 new customers if there were no sales in 2022?
 2. Our customer acquisition for 2025 was horrible and dropped off severely.
 3. We have no new customers in 2026 either. Why?
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Measure Exploration

- We have 60 products in our line-up (models)
 - There are 3 sales channels
 - There are 6 stores (including an "online" warehouse)
 - There are 1,000 orders in total across the date range for the data
 - Overall sales are *extremely* low for a large-scale company
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Magnitude Analysis

- The accessories that generate the most sales by far are gloves
- The bike that sells the most is the road bike
- The total quantity of bike sold is < 700 for the entire date range (combination of all 3 bike types)
- Gloves and helmets are the top sellers (gloves 446 pairs sold)
- Bikes are premium bikes with the cheapest avg. price (mountain bike) at \$1550 and most expensive (hybrid) close to \$2000
- Gloves and bike lights have the highest average prices in accessories
- Texas, Florida and Carolina are the states with the most customers
- The sales channel with the highest sales revenue is "Dealer". The lowest is "Store"
- All sales channels are selling approximately the same quantity
- All stores have approximately the same sales revenue and quantity sold.

NOTE: The data is not realistic (it was generated by ChatGPT); accessories such as gloves and helmets cost more than bikes.

- Although the data is not realistic, it is still usable and compatible for data model building.
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Ranking Analysis

- The bike with the top profit was a mountain bike > Terra-228 > \$1,043
- The accessory with the top profit was a helmet > AirFlow-29 > \$938
- Our top two customers are: Harper Clark and Grace Clark w/ approximately \$55k each
- Below are the top models for each category based on profit:

	category	↑↓	🔍	model_name	↑↓	🔍	profit	↑↓	🔍	top_profit_products	↑↓	🔍
1	Bike Light			Core-71			918.710000			1		
2	Bike Pump			AirFlow-38			580.421111			1		
3	Gloves			Lumen-95			917.511176			1		
4	Helmet			AirFlow-29			938.311428			1		
5	Hybrid Bike			Pulse-548			957.922142			1		
6	Mountain Bike			Terra-228			1043.820526			1		
7	Road Bike			Pulse-820			939.208500			1		
8	Water Bottle			Hydro-74			897.158823			1		

EDA Part 2

Cumulative Analysis

- Moving average increased significantly for the first two quarters of 2023 and then more or less stabilized.
 - This shows little to no improvement in revenue YoY and a lack of growth in sales.

Performance Analysis

- We compiled the data to use in our Dashboard model for comparing the current period (year) revenue to the previous year's revenue & the average sales revenue for all of our products.

Part to Whole Analysis

- Compared the total sales revenue for all of our categories as a percentage of total overall revenue.
- Made the shocking discovery that the highest contributor to sales revenue (almost doubling second place) were "gloves", an accessory, and not actually a bike (our main product).

Data Segmentation Analysis

- We have:
 - VIP: 90 customers
 - Plus: 28 customers
 - Regular: 2 customers
- How many models are categorized as "high" profit margin categories? 35
- How many models that are "high" profit margin categories are active? 27
- How many models that are "high" profit margin categories are inactive? 8